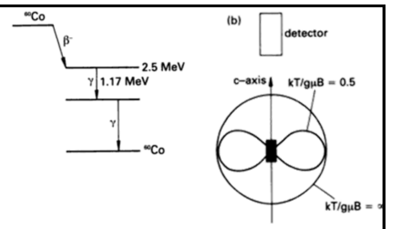
# Temperature sensors

**Primary thermometers:** Temperature derived without unknown quantities. E.g.:

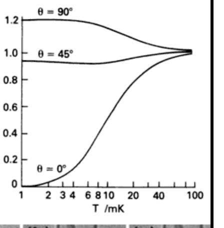
- From thermodynamic properties, such as **sound velocity in noble gases** (see metrology slides)

- **Johnson-Nyquist noise** (see metrology slides)

- **Anisotropic angular distribution of gamma radiation** for a radioactive isotope in B space. Principle: the magnetic moment of a $^{60}Co$ atom is polarized in B field. In a ferromagnetic $^{59}Co$ background, the field felt by $^{60}Co$ is B=10T . Its nuclear spin undergoes hyperfine splitting to N different levels, $\Delta E \sim NB$, where $\Delta E/k_B$ = 31mK → thus in the mK range the probability of filling different levels is strongly T dependent. At low T ($k_B T << \Delta E$) only the lowest level is occupied whose gamma decay has a temperature dependent anisotropic angular distribution that can be measured. Sensitivity: good for the mK range, typical calibration procedure for dilution refrigerators.




*(top left) decay process of $^{60}Co$, (top right) angular distribution of the gamma decay with respect to the B field direction (bottom) radiation intensity as a function of angle and temperature*

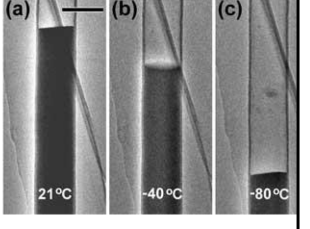



**Secondary thermometers:** A change in some property of a sensor that is being measured (e.g. resistance, pressure, capacitance). Calibration is required. They usually give a more convenient, more accurately measured signal.

**Types:** -Thermal expansion based

-Thermocouples

-Resistance thermometers , Resistance temperature detectors (RTD), Thermistors,…




*A Liquid-Ga-Filled Carbon Nanotube: A Miniaturized Temperature Sensor, Small, 1, 11, 1088 (2005).*

*http://en.wikipedia.org/wiki/Resistance_thermometer*,
http://www.elprocus.com/temperature-sensors-types-working-operation/

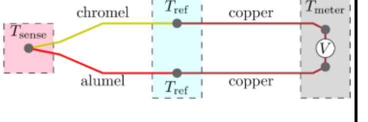


# Temperature sensors: thermocouples

**Thermocouples**

Principle: When two conductors of different materials come into contact, a so-called contact voltage is generated. In a chain of conductors using different materials, these contact voltages are cancelled if (i) the two ends are made of the same material (e.g. copper) and (ii) all connection points have the same temperature. However, the contact voltage is temperature-dependent, i.e. a net thermoelectric voltage is measured if one connection point is at the temperature under test ($T_{sense}$) and the other connection points are at a well-defined reference temperature ($T_{ref}$). The measured thermoelectric voltage is proportional to the temperature difference, V=S* ($T_{sense}$ -$T_{ref}$). The phenomenon is based on the Seebeck effect (1821), the sensitivity being determined by the Seebeck coefficient S.

*Operation principle of a so-called K-type thermocouple using chromel and alumel wires, i.e. two different alloys the first containing chromium and the last aluminum. This thermocouple has a 41μV/K sensitivity.*




$$\nabla V = -S(T)\nabla T,$$

Advantages: cheap, no control voltage required, can be used up to very high temperatures (~2300°C ), where components of other, more complex thermometers would be damaged.

Disadvantages: inaccurate, (≈1C°) the sensitivity is small (<80 μV/°C, see characteristics in the right)

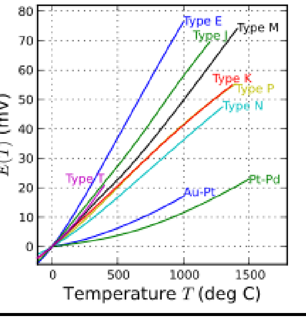
Example: type E thermocouple (68 μV/°C ) using Nickel and Chromium-Constantan,

Selection criteria for the thermocouple wires: sensitivity, price, melting point, chemical reactivity

Application: flame sensors, oven thermometers, engine thermometers

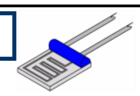
*Typical thermocouple thermometer*

*Response of thermocouples as a function of ΔT*




http://en.wikipedia.org/wiki/Thermocouple

## Temperature sensors: metallic resistance temperature detectors (RTD)



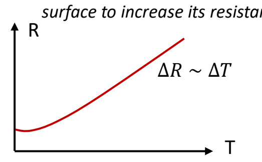
*Resistance thermometer: the thin film Pt wire makes many turns on the sensor surface to increase its resistance*

Principle: The temperature is derived by measuring the change in resistance of the sensor (usually metal). Most widely used temperature sensor.

Advantage: Simple principle, easy to measure, low drift, good stability.

Disadvantage: low sensitivity (e.g. ΔR/R/ΔT ~ 0.4%/C° for Pt), slow response.

Metals used: Pt (stable resistance-temp. relationship over a wide T range), Ni (non-linear up to about 300 C°), Cu (most linear, but can oxidize above 150 C°)




*In a metallic RTD the resistance change is proportional to the temperature change over a broad temperature interval, except for the very low temperatures, where a saturated remaining resistance is observed*

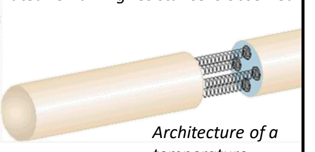
It reproduces so well that it is also used as a temperature measurement standard. See SPRT (standard platinum resistance thermometers) ITS-90.

To obtain a wll-measurable resistance:

Thin film (1-10nm thick). Thermal expansion difference between substrate/metal results in stress → often a coiled wire is used in a tube filled with gas or ceramic powder to ensure free movement in case of thermal expansion. ITS-90 is also based on this. E.g. in ITS-90 (International Temperature Scales 1990) the Pt SPTR is recommended to be used over a wide T range, see Wiki table.

Sensitivity for a Pt100 sensor: 0.385 Ω/°C. To measure such a small resistance variation, a four probe measurement method is required (see the slides of the first lecture)

What exactly are we measuring the temperature of? E.g. circuit vs. environmental temperature in a mobile phone, or lattice vs. electron temperature in cryogenic meas.



*Architecture of a temperature sensor based on coiled Pt wire*

Standard interpolating thermometers and their ranges [edit]

| Lower (K) | Upper (K) | Variations | Thermometer | Calibration and interpolation strategy |
|---|---|---|---|---|
| 0.65 | 3.2 | 1 | Helium-3 vapor pressure thermometer | Vapor pressure–temperature relationship fixed by a specified function. |
| 1.25 | 2.1768 | 1 | Helium-4 vapor pressure thermometer | Vapor pressure–temperature relationship fixed by a specified function. |
| 13.8033 | 1234.93 | 11 | Platinum resistance thermometer | Resistance calibrated at various fixed points and interpolated in a specified way. Eleven distinct calibration procedures are specified. |

http://en.wikipedia.org/wiki/Resistance_thermometer



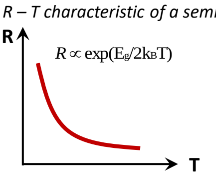
## Temperature sensors: semiconductor thermometers

**Thermistor**

It works in a similar way to RTDs but measures semiconductor resistance, which is highly non-linear (see figure). High sensitivity over a small range.

Applications: overheating inhibitors, working-point applications, where a small temp. range should be investigated with high sensitivity, temp. measurement at very low temperatures, where metallic RTDs are already saturated, …

*R – T characteristic of a semiconductor thermometer*

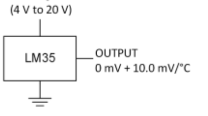



Semiconductor IC thermometers

Advantage: sensor is integrated to the circuit, includes ADC, programmable, sufficiently large output signal, …

E.g. Texas Instruments LM35 - 1.5$/piece

*Texas Instruments LM35 IC thermometer*



## Light sensors

Light is a small segment of EM waves: a range that can be perceived by the eye.
 c=f λ, λ= 400-700nm range

Particle nature of light (Einstein 1905, photoelectric effect)
 Photon: light transports energy in quanta of E=hf, where h is the Planck's constant.

Wave nature of light: it behaves as a wave according to Maxwell's equations

The particle – wave duality of light is exploited by sensors, some of which are based on the particle nature (photodiodes); some exploit the wave nature (bolometers).

**Photodiode**

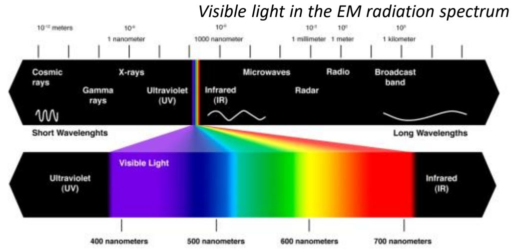
Semiconductor diode in PN or PIN junction (I insulator) with reverse bias

Principle (photoelectric effect): iIf the energy of the incoming photon is greater than the band gap (i.e. E>Eg), it excites electrons from the valence band band into the conduction band. The applied voltage V causes the electrons to leave the electrons towards the N-doped cathode side and the holes towards the P-doped anode, generating a photocurrent.

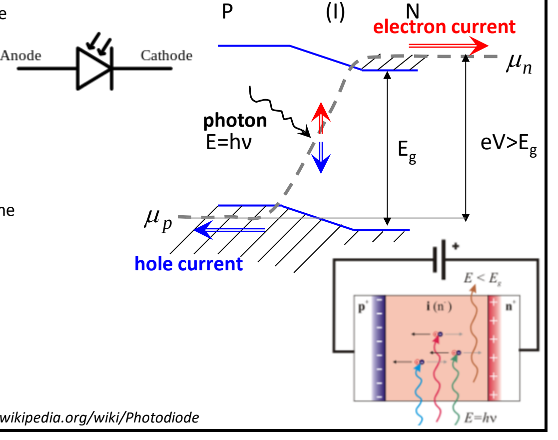
Biasing increases response speed (RC term) but also noise background (dark current).

Applications: CD readers, remote controls, optocouplers, optical communication, ....

*Visible light in the EM radiation spectrum*




*Photodiode circuit diagram and operating principle*




*Blackburn, Modern Instrumentation for Scientists and Engineers, http://en.wikipedia.org/wiki/Photodiode*

25

## Light sensors

**CCD (Charge Coupled Device)**

1969 Bell Labs, Boyle and Smith (2009 Nobel prize)

Principle: Light creates trapped electrons in a MOS capacitor array, where each capacitor is a pixel of the sensor. Then, using the pixels as a shift register, the trapped electron can be delivered to the edge of the sensor, where a charge amplifier can be used to convert each pixel signal into a voltage signal and store it one by one.

**CMOS Active Pixel sensors**

Each pixel contains a photodiode (PPD, pinned photodiode, p+/n/p junctions) and an active amplifier.

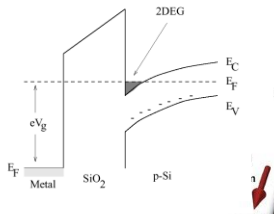
Mobilephone cameras, webcams and digital cameras after 2010 use this technology.

Advantages: cheaper, lower power consumption than CCD, easier fabrication, faster readout, can do both sensing and processing, less crosstalk to neighbouring pixels than in CCD.

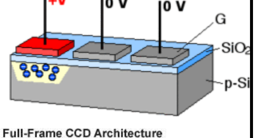
**Further light sensors:**

 Photoresistor, avalanche photodiode single photon detector, photomultiplier tube, ...

*MOS (Metal /oxide / semiconductor) band structure with inversion layer on the surface. The potential well can trap electrons*




*Charge coupled principle, as the trapped charges are delivered pixel by pixel to the amplifier*




Full-Frame CCD Architecture

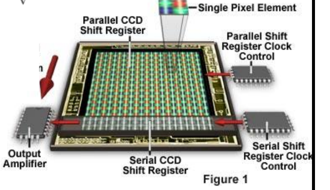



Figure 1

Anatomy of the Active Pixel Sensor Photodiode

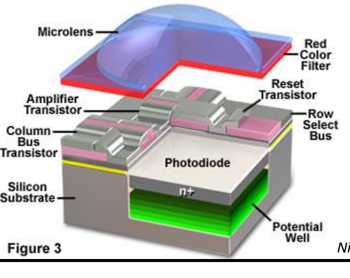
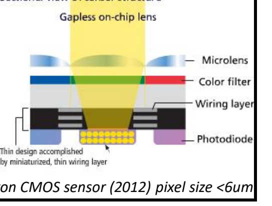



Figure 3




*Nikon CMOS sensor (2012) pixel size <6um*

26

## Sensing electromagnetic radiation: bolometers

Principle: It contains an absorption element (e.g. a thin layer of metal) which, when exposed to radiation, increases the temperature which is measured. It is sensitive to the energy left in the detector. The detector is usually connected to a cooling tank.

1880 Langley first bolometer. Platinum band, Wheatstone bridge arrangement. He could detect 1 cow at half a mile.

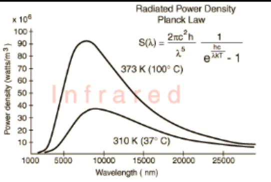
Used in the infrared and terahertz range ($\lambda$= µm - mm).

Application: astronomy, particle physics (cooled detectors), Thermal imaging cameras, ...

Mikrobolometer (in thermal imaging cameras):

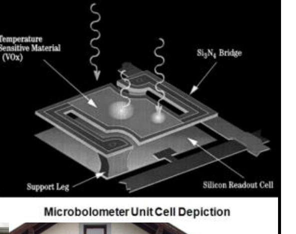
The sensor is built with resistance measurement electronics integrated in a silicon chip and a suspended vanadium oxide or amorphous silicon wafer above (~1µm), which varies its resistance due to the irradiation. Suspension provides thermal insulation from the chip. Sometimes a heat mirror is put on the Si substrate (e.g. titanium) under the wafer. Typical resolution: 1024x768

Uses: thermal engineering, medical diagnostics, military, security engineering, automotive (animal detection), ...

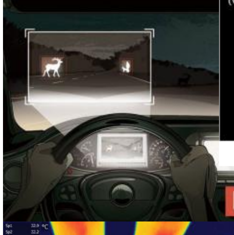


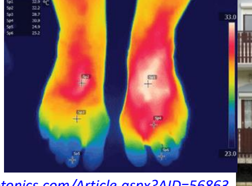

*Black body radiation at different temperatures. At 37 C° maximum intensity around 10µm wavelength.*

*Microbolometer design: suspended VOx plate above the chip*




*Autoliv: night time wildlife detection, signals and spot light illumination. Mercedes S-Class (2014)*





*(left) Thermal camera at airport to detect fever from H1N1 virus,*
*(right) Diagnosing inflammation due to increased blood flow*
*(right 2) Thermal camera image of apartment building in winter.*

http://www.photonics.com/Article.aspx?AID=56863,
http://en.wikipedia.org/wiki/Microbolometer



## Microelectromechanical systems (MEMS) as acceleration sensors

**Acceleration sensor:**

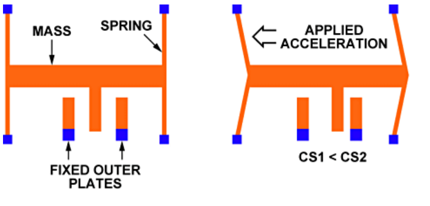
Principle: A fixed plate provides one electrode of a capacitor, the other electrode is suspended on micromachined springs, having a well-defined mass. When the sensor accelerates in x direction, the mass is displaced by the amount determined by the springs. Due to the displacement, the capacitance between the two electrodes. On the figure two fixed electrodes are used, CS1 capacitance decreases and C2 increases due to the displacement.

Silicon-based devices (good mechanical properties, rigid ), can be made using IC manufacturing techniques. Suspended structures are created over a silicon chip (using a sacrificial layer)

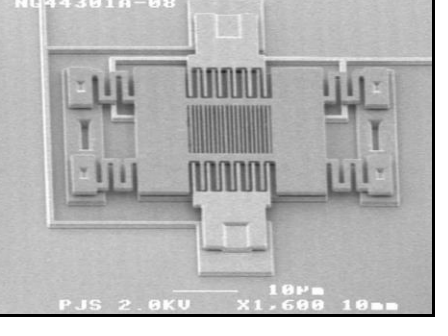
Pl. ADXL321 4mmx4mm, resolution: 3mg@50Hz, range: ±18g, X &Y axis

Static acceleration ("g") and dynamic acceleration are both measured. Price: 8$.

*Operation of MEMS accelerometers: Capacitive measurement of the displacement of a mass on a spring*




*SEM image of accelerometer with mass suspended on springs in the middle, and capacitive readout at the sides.*



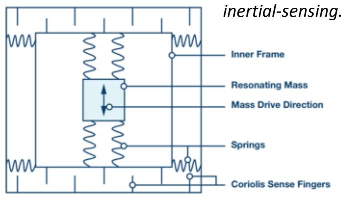

*Blackburn, Modern Instrumentation for Scientists and Engineers,*
*http://www.memsjournal.com/2011/01/motion-sensing-in-the-iphone-4-mems-gyroscope.html*
*http://www.digikey.com/en/articles/techzone/2011/apr/mems-accelerometers-gyroscopes-and-geomagnetic-sensors---propelling-disruptive-consumer-applications*

## MEMS gyroscopes

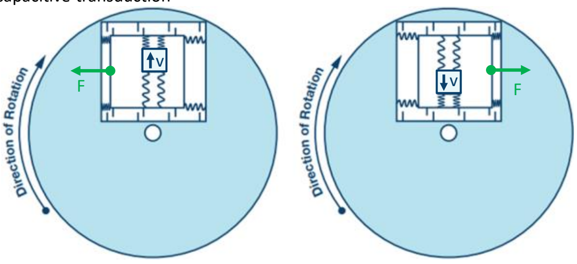
*Source of text and illustrations: https://www.analog.com/en/resources/technical-articles/mems-gyroscope-provides-precision-inertial-sensing.html*

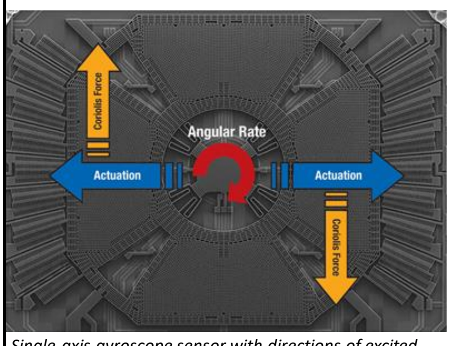
Principle: The Coriolis force on a resonating mass is proportional to the angular velocity ($\boldsymbol{F}_C = -2\text{m}\boldsymbol{\Omega} \times \boldsymbol{v}$).

When the resonating mass moves toward the outer edge of the rotation, it is accelerated to the right and exerts on the frame a reaction force to the left. When it moves toward the center of the rotation it exerts a force to the right, as indicated by the green arrows. The frame containing the resonating mass is tethered to the substrate by springs at 90° relative to the resonating motion, as shown in the Figure. This figure also shows the Coriolis sense fingers that are used to sense displacement of the frame through capacitive transduction in response to the force exerted by the mass.




E.g. ST L3G4200D, 3axis digital output, size: 4x4x1mm, Sensitivity 8m°/sec, max range 2000 °/sec




*Single-axis gyroscope sensor with directions of excited vibration, rotation and Coriolis force,* STMicroelectronics

*Blackburn, Modern Instrumentation for Scientists and Engineers, http://www.memsjournal.com/2011/01/motion-sensing-in-the-iphone-4-mems-gyroscope.html*
*http://www.digikey.com/en/articles/techzone/2011/apr/mems-accelerometers-gyroscopes-and-geomagnetic-sensors---propelling-disruptive-consumer-applications*

29

## MEMS acceleration sensors and gyroscopes

Usage:

Engine acceleration or vibration, airbag controls (collision, upside down), corner stabilizers, ESC (electronic stability control), braking system activation in case of sudden change of direction, Automatic collision notification (Toyota, Lexus, BMW, Ford, ...)

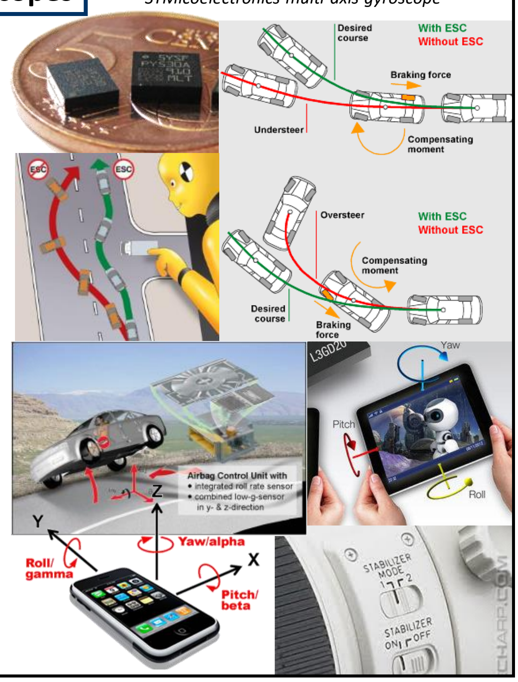
Earthquake forecasting, building stability testing, navigation systems (in the absence of GPS)

Orientation of mobile devices (standing/lying), FFS (free fall sensor) in notebook hard disks (Dell, Lenovo, Apple, WD ...) - picks up the reading head from the disk. Nintendo playstation - hand movement tracking.

Image stabilisation for cameras

Gravitational field meters

*STMicoelectronics multi-axis gyroscope*




30

# Piezoelectric acceleration sensors

Piezoelectricity: Mechanical stress → deformation → surface charge appears on piezoelectric material.

E.g. in the case of quartz ($SiO_2$), due to the crystalline arrangement of the atoms as shown in the figure (see quartz clocks, ultrasonic generators, etc.)

Acceleration can be converted to force measurement. At constant force, the charge leaks (leakage resistance), so it is not suitable for measuring dc force. In ac, it can operate up to high frequencies.

Types:

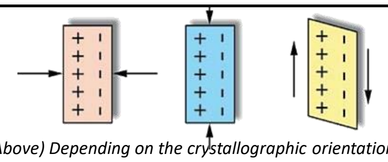
a) Fixed mass on a piezo block

b) Shear piezo, mass is on the side of the piezo.

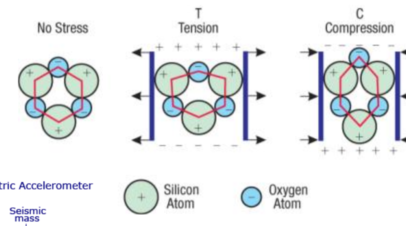
Typical parameters:

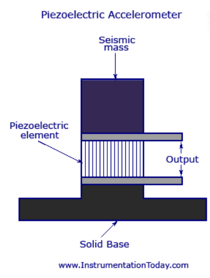
Range can reach 10000g, 1Hz - >10kHz, 10-100mV/g

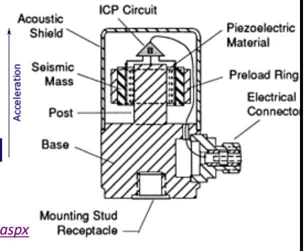
Use: in shock and collision sensors



*(Above) Depending on the crystallographic orientation of the quartz, different piezoelectric phenomena can be observed (A) longitudinal, (B) transverse, (C) shear. (Below) Microscopic origin of the transverse case.*




*Accelerometers structure: inertial mass attached to the piezoelectric material with longitudinal piezoelectric effect on the left and shear piezoelectric effect on the right.*

*Blackburn, Modern Instrumentation for Scientists and Engineers, [https://www.pcb.com/TechSupport/Tech_Accel.aspx](https://www.pcb.com/TechSupport/Tech_Accel.aspx)*
*http://www.sensorsmag.com/sensors/acceleration-vibration/the-principles-piezoelectric-accelerometers-1022*